

## Machine Learning(CAI-262)

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- Terms used in Machine Learning
  - Steps involved in training Machine learning Model
  - Feature scaling and normalization
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### Terms used in machine Learning

Machine learning is a part of artificial intelligence that allows models to continuously learn by using past experiences, exploring data, and identifying patterns with little human intervention.

Machine learning (ML) teaches computers to learn from data to make accurate predictions or decisions.

Imagine teaching a child to recognize and name fruits – an apple, for example. You show them several pictures of apples, and each time, you say, “This is an apple.” The child sees various apples – green ones, red ones, big ones, and small ones. They hear the word “apple” associated with the picture each time. Eventually, the child starts to identify what characteristics make an apple – perhaps the round shape, the stem at the top, or a particular range of colours.

Now, when you show the child a new picture of an apple, even one they’ve never seen before, they identify it correctly. Why? Because through repeated examples (the training), they’ve learned to associate specific features (shape, colour, stem) with the label “apple.”

In machine learning, this process is akin to supervised learning.

- The child is the **algorithm**.
- The pictures of apples are the **data**.
- Your repeated saying, “This is an apple,” is the **labelling**.
- The model training is the child’s learning process, identifying features like shape and colour.
- The child’s ability to name a new fruit correctly is the **prediction**.

**Data:** Any material which is being used for learning purpose is data, it can be in the form of images, audios, videos, text etc.

**Sample:** The data we use for representing one example is called a sample.

**Features:** The characteristics of one object which makes it different from other objects are called features. (features should be in numeric form)

**Training / Testing data:** To make a machine do learn some data we divide the dataset into two parts, training dataset (to make machine learn data) and testing dataset (to check/test whether machine learnt something or not).

**Performance Evaluation:** Testing the performance of a model whether it had learnt something or not for which test data set is used. The samples in test dataset are unknown for model as it was not trained on them in this way we can see how well it perform on unseen samples.

**Performance Metrics:** An amount which show how good our trained model is on test dataset. Accuracy is a performance metric which tells the performance of a model.

$$Accuracy = \frac{\text{Correct predictions}}{\text{total test samples}} * 100$$

**Label/Class:** Type of samples present in a dataset e.g Trucks and cars, nine digits of numbers etc.

**Predicted label:** The class an ML model is predicting after getting trained (represented as  $\hat{y}$ ).

**Actual Label:** the actual class a sample belongs to (represented as  $y$ ).

## Steps involved in training Machine learning Model

### 1) Data Pre -Processing

- Collect the data

- Clean the data

- Feature Scaling/ normalization.

- Split into training & test sets.

### 2) Modelling

Build the model

Train the model (on training data set).

Make predictions (on test data set).

### 3) Evaluation

Calculate performance metrics.

Make a verdict.

## Data Pre-Processing:

This step is very important, as in Machine learning the process of learning purely depends upon the quality of data. After collection of data and its cleaning a very crucial step is scaling and normalization of data.

Data preprocessing is a process of preparing the raw data and making it suitable for a machine learning model. It is the first and crucial step while creating a machine learning model.

When creating a machine learning project, it is not always a case that we come across the clean and formatted data. And while doing any operation with data, it is mandatory to clean it and put in a formatted way. So for this, we use data preprocessing task.

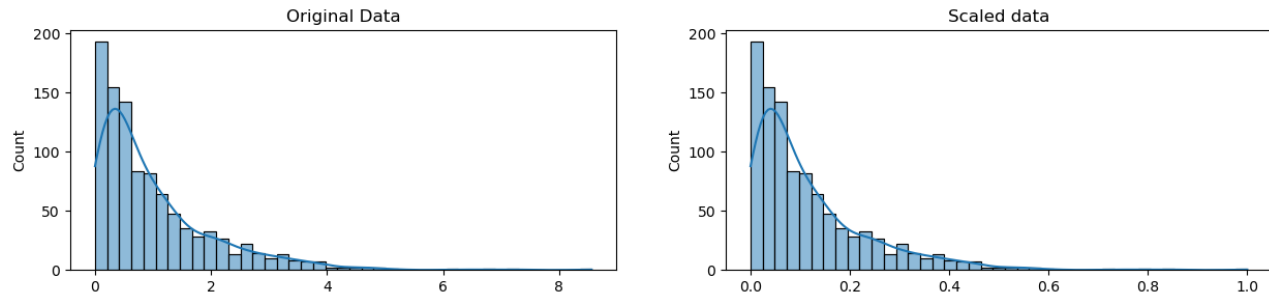
## Scaling/ normalization of data:

In scaling/normalizing we transform the values of numeric variables so that the transformed data points have specific helpful properties. The difference is given below.

- in **scaling**, we change the range of our data, while
- in **normalization**, we change the shape of the distribution of our data.

## Scaling:

This means that we are transforming our data so that it fits within a specific scale, like 0-100 or 0-1. We want to scale data when we are using methods based on measures of how far apart data points are, like support vector machines (SVM) or k-nearest neighbors (KNN). With these algorithms, a change of "1" in any numeric feature is given the same importance.



Notice that the *shape* of the data doesn't change, but that instead of ranging from 0 to 8, it now ranges from 0 to 1.

One formula for scaling data is given below

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Consider the dataset given below if we try to scale our dataset using above formula then data will change as follows.

| Salary | age |
|--------|-----|
| 70000  | 45  |
| 60000  | 44  |
| 52000  | 40  |

After scaling dataset would be in range [0 ; 1]

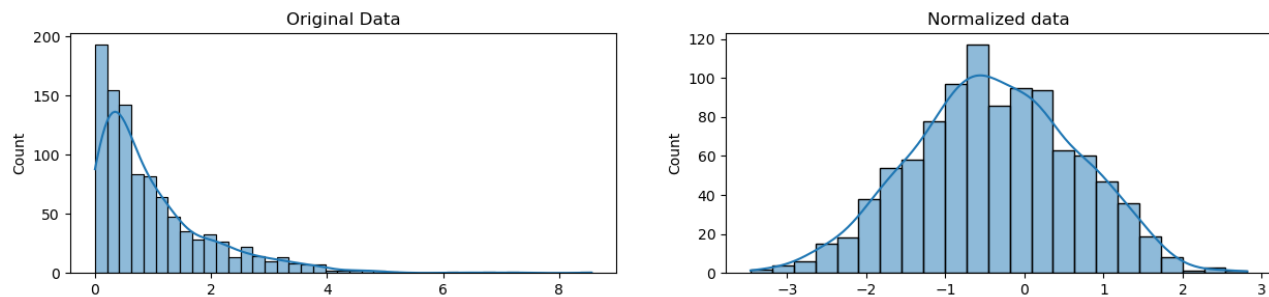
| Salary | age  |
|--------|------|
| 1      | 1    |
| 0.44   | 0.75 |
| 0      | 0    |

## Normalization:

Scaling just changes the range of your data but normalization change our observations so that we can be described as a normal distribution.

*Normal distribution: Also known as the "bell curve", this is a specific statistical distribution where a roughly equal observations fall above and below the mean, the mean and the median are the same, and there are more observations closer to the mean. The normal distribution is also known as the Gaussian distribution.*

In general, we normalize our data if we are going to be using a machine learning or statistics technique that assumes your data is normally distributed. Some examples of these include linear discriminant analysis (LDA) and Gaussian naive Bayes.



Notice that the *shape* of our data has changed. Before normalizing it was almost L-shaped. But after normalizing it looks more like the outline of a bell (hence "bell curve").

The formula for normalization is given below.

$$X' = \frac{X - \mu}{\sigma}$$

## Data Splitting:

We need to split a dataset in order to perform unbiased model evaluation and identify underfitting or overfitting. Data splitting is when data is divided into two or more subsets.

### 1) Train/Test split:

Typically, with a two-part split, one part is used to evaluate or test the data and the other to train the model.



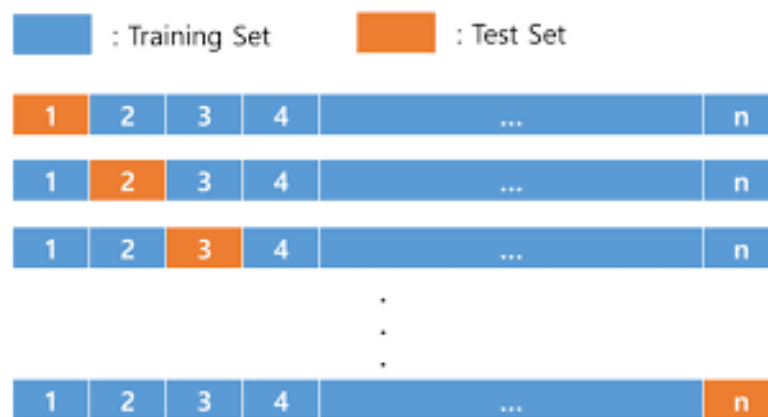
**Training Set:** A subset of dataset to train the machine learning model, and we already know the output.

**Test set:** A subset of dataset to test the machine learning model, and by using the test set, model predicts the output.

In train/test split the portion of training set is always larger than test set usually 70/30, 75/25, 80/20, 78/23 etc. Use this strategy when a dataset is very large.

## 2) Leave One Out Cross validation:

Leave One Out Cross Validation (LOOCV) is used when a dataset is very small and we don't have enough data to test the performance of the model. In this strategy only one sample is kept "out" for testing and model is trained on rest of all other samples (i.e.  $n-1$ ). Then check the prediction of model on that one sample. Repeat this process  $n$  times, thus  $n$  models are trained and all  $n$  samples are tested one by one. This is computationally expensive.



## 3) K-Fold Cross Validation:

In K-Fold Cross Validation, we split the dataset into  $k$  number of subsets (known as folds) then we perform training on the all the subsets but leave one subset for the evaluation of the trained model. In this method, we iterate  $k$  times with a different subset reserved for testing purpose each time.

Following is an example of 5-fold.

